Rogue Merchant: A Dockerized Fantasy Bazaar

Buy, sell, and trade fantasy goods across 10 market rounds. Prices fluctuate server-side—containerized with Nginx, Flask, and MySQL.

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The Deployment Nightmare

Python Version Mismatch

Local vs. production interpreter conflicts break everything

MySQL Setup Pain

Manual installations, schema seeding, encoding issues

Nginx Configuration

Reverse proxy rules, port conflicts, permission headaches

It Works On My Machine

Environment drift kills reproducibility and collaboration

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Enter Docker: The Container Solution

Package code + dependencies + runtime into isolated, portable units. One command deploys the entire stack.

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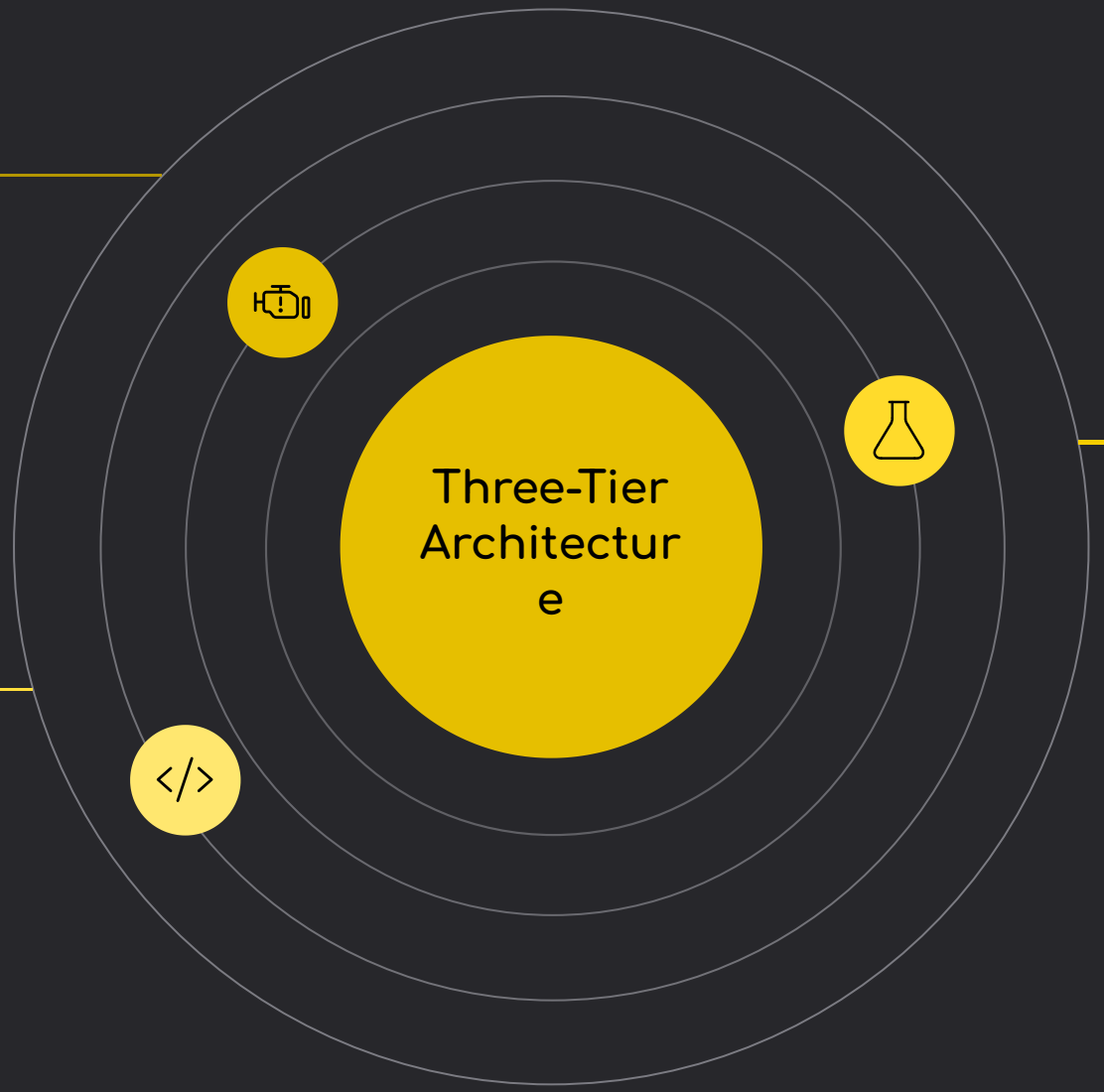
Three-Tier Architecture

Nginx Frontend

Serves static files and proxies API requests

MySQL Database

Provides persistent relational data storage



Flask API

Handles business logic and state management

01

Nginx

Frontend static files + API proxy

02

Flask API

Business logic, state management

03

MySQL

Persistent data storage

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Multi-Stage Builds: Backend Efficiency

Builder Stage

Heavy build tools + compile Python dependencies into virtual environment

Runner Stage

Copy clean environment only → tiny, secure production image

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Docker Networks: Secure Internal Communication

Nginx

Routes to backend:5000 via Docker DNS

Isolated Bridge

rogue-net keeps ports internal

Flask

No host exposure needed

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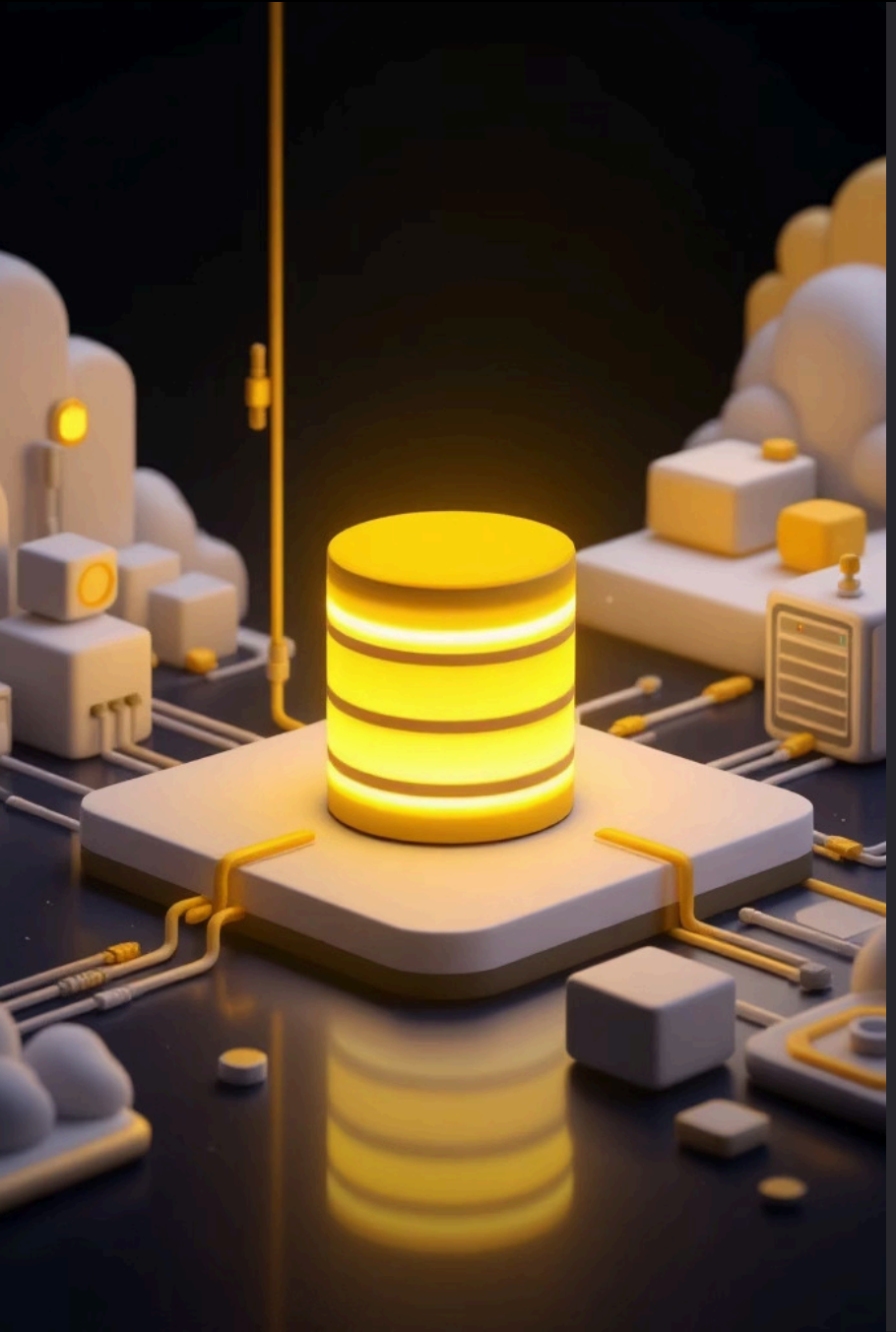

Docker Volumes: Data Persistence

Containers Are Ephemeral

Restart = data loss without volumes

Volume Mapping

db_data survives crashes, recreations

Player State

Gold, inventory, transactions preserved

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Production-Ready Commands

Launch Everything

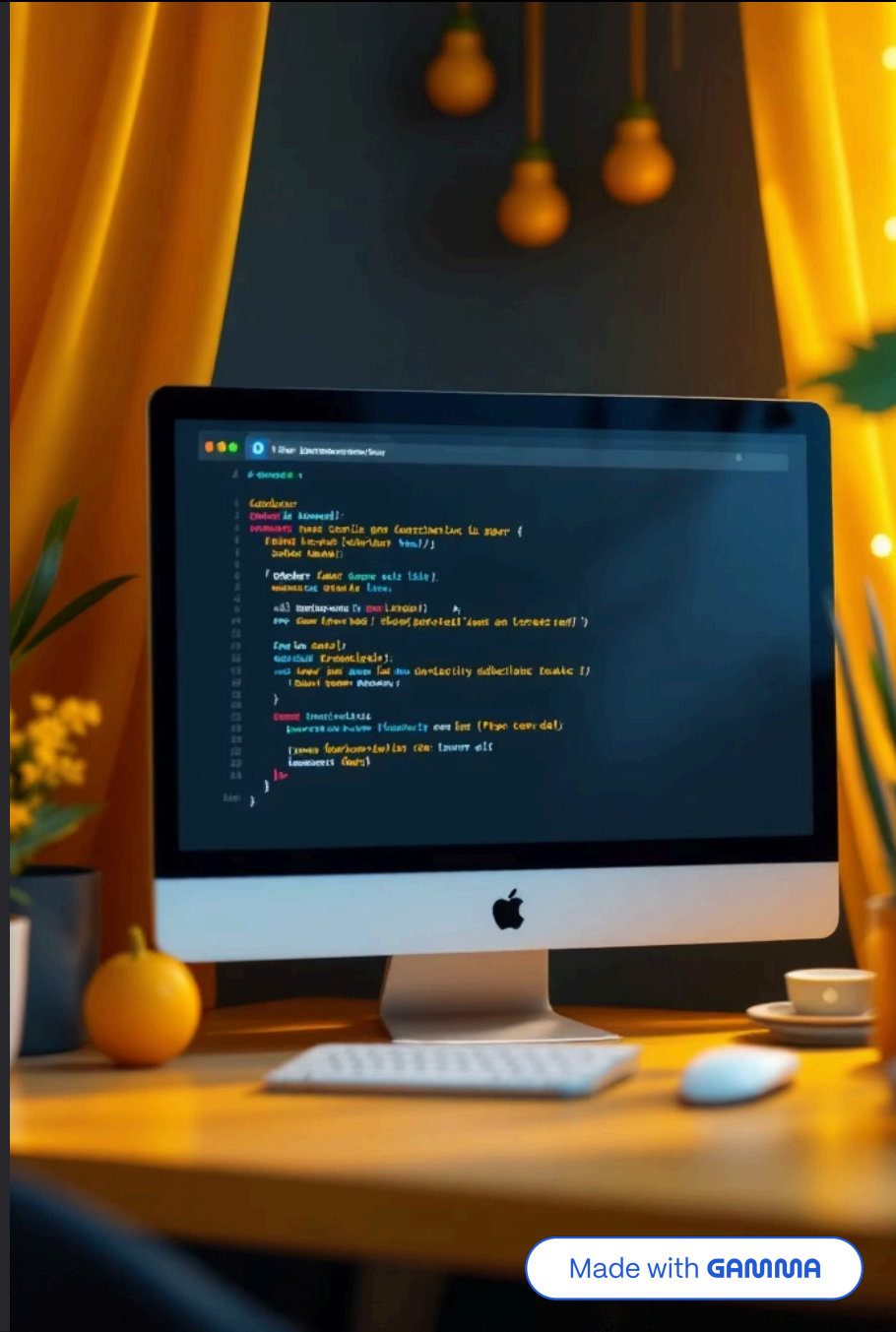
```
docker compose up -d --build
```

Build images, start containers, seed schema automatically

Wipe Data Clean

```
docker compose down -v
```

Destroy containers + volume for fresh start



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Key Benefits of Docker Compose

Layered Images

Multi-stage builds optimize size and security

Isolated Networking

Internal DNS eliminates port conflicts

Persistent Storage

Volumes protect critical game state

Quick Iteration

Build once, deploy anywhere, scale easily

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Ready to Containerize?

Clone the Repo

docker compose up

Play & Contribute

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